

SIH26034: Proposed Feature Evaluation

Discussion draft for faculty and team — feature selection only; not a task allocation or development timeline.

Project Context

SIH26034 concerns software that scans packaged-commodity labels, images and product information; extracts prescribed declarations; and assists Legal Metrology officers in assessing potential compliance issues. The baseline is OCR plus deterministic rule checks. Our proposed set adds field-workflow value: traceable evidence, calibrated measurement assistance, package-history comparison and transparent inspection prioritisation.

Positioning: report “**potential non-compliance**”, evidence, confidence and the need for officer verification. The system must not claim to make final legal findings or verify actual contents/weight from an image.

Legal Rules: Scope for Prototype

The table below is a **prototype-oriented legal checklist**, not a substitute for the consolidated Rules or legal advice. The Legal Metrology (Packaged Commodities) Rules, 2011 must be read together with amendments currently in force. Before final implementation, validate every rule text, schedule, exemption and commencement date against official Department of Consumer Affairs/Gazette materials.

Core statutory context: Section 18 of the Legal Metrology Act, 2009 prohibits manufacture, packing, import, sale, distribution, delivery or offer/exposure for sale of pre-packaged commodities unless prescribed declarations are made on the package in the prescribed manner.

Rule / provision	Prototype check	What the system can do	Legal/technical boundary
Legal Metrology Act, 2009, s.18	Pre-packaged commodities must bear prescribed declarations in prescribed manner.	Use as overarching authority in report header and rule-engine source metadata.	System supports inspection; it does not itself determine liability.
LMPC Rules, 2011, Rule 6(1)(a)	Name and address of manufacturer / packer / importer, as applicable.	OCR, keyword detection, address-pattern extraction and missing-field flag.	OCR may miss text; address sufficiency and responsible party applicability require officer review.
Rule 6(1)(b)	Common or generic name of commodity.	Extract product/generic-name candidates; compare with product category/label context.	Generic name can be context-dependent; do not auto-conclude absence from a partial image.
Rule 6(1)(c)	Net quantity in standard unit of weight/measure or number, as applicable.	Extract value and unit; validate syntax/allowed units; highlight declaration crop.	Image can check declared value/format only—not actual contents or net quantity. Physical weighing/measurement is separate.
Rule 6(1)(d)	Month and year of manufacture, packing or import, as applicable.	OCR/date parsing; flag missing or unparseable text.	Date applicability and acceptable declaration variant may depend on product/rule context; officer review needed.
Rule 6(1)(e)	Maximum retail price inclusive of all taxes; consumer-care details as prescribed.	Extract MRP/Rs./INR amount; detect tax-inclusion wording; detect phone/email/contact candidates.	Support multiple lawful wordings; validate with current official rule text. Cannot establish whether price charged at sale exceeded MRP from label alone.
Rule 6(1)(f)	Relevant size/dimension/other declaration where applicable.	Detect dimension patterns and flag context-dependent absence for review.	Not universally required; applicability is conditional. Never treat absence as automatic violation without category/context assessment.
Rule 6(10)	E-commerce display of applicable mandatory declarations; manufacture/packing/import month/year has historically been treated differently for online display.	For an authorised/static listing, check visible listing text/images against configured online declaration checklist.	Verify current wording/effective status before coding. Live scraping can be restricted by platform terms and technical controls.
Rule 6(10A), if in force / applicable	Recent e-commerce / imported-product country-of-origin functionality requirements require official verification of final text and commencement.	Treat as optional, versioned e-commerce check: COO displayed and, if applicable, filter capability observed.	Do not claim enforcement status until official notification and effective date are verified. A screenshot cannot always prove dynamic filter functionality.

Rule / provision	Prototype check	What the system can do	Legal/technical boundary
Rules 7–8 and relevant schedules	Manner, legibility, placement / principal-display-panel and prescribed character/letter/numeral size requirements, subject to current text and exemptions.	Image-quality and visibility flag; calibrated font-height <i>estimate</i> ; evidence crop and location overlay.	An uncalibrated image cannot give physical mm. A calibrated estimate remains decision-support; physical measurement is required for enforcement-grade finding.
Rules 9–13 and relevant schedules	Package-specific declaration, unit and quantity-expression requirements; applicability can depend on commodity and package type.	Versioned rule-engine checks for standardised units, patterns and category-specific fields after legal validation.	Do not hard-code a generic checklist as universal. Exemptions, schedules and category-specific rules must be confirmed.
Rule 15 and other category-specific provisions, where applicable	Unit-sale-price / other product-specific requirements, if applicable.	Calculate/display possible unit-price consistency and flag for officer review.	Applicability must be legally configured by product category and current notification; avoid universal automatic flagging.

Key Legal Controls in the System

1. Store each rule as versioned data: rule number, authoritative source URL/Gazette number, effective-from date, category applicability, exception/exemption and validation logic.
2. Separate **presence/format detection** from **legal conclusion**: “not detected in submitted image” is not the same as “legally absent.”
3. For font size, package area, placement, readability and category-specific conditions, show a confidence and a mandatory officer-review action.
4. Use only officially verified legal text for final reports; avoid copying rule summaries from blogs or LLM outputs.

Feature 1 — Evidence-First Inspection Record

Feature & purpose. Preserve original photo, SHA-256 hash, timestamp, optional location, OCR output, extracted field, rule result and officer verification in one traceable record. A hash is a file fingerprint: a changed file produces a different hash.

Coverage. Supporting photographs, reports, inspection repository/history, secure workflow and dashboard.

Differentiation. Generic tools show a result. This records *image* → *OCR* → *field* → *rule* → *officer decision* → *report*.

Pros / limitations. Low-cost, auditable and highly demoable; but a hash proves file integrity after capture, not that the original photo or claimed location is true.

Risks. Preserve originals; maintain transformation metadata; minimise access to location/personnel data; use encrypted storage and role-based access.

Difficulty / feasibility. Easy; highly feasible.

Legal/regulatory care. Electronic-evidence procedure and certification requirements must be validated for the intended proceeding; do not state that a hash alone makes evidence court-admissible. Follow DPDP Act, 2023 principles for personal/location data.

Technology. Python hashlib / Node crypto, PostgreSQL or SQLite, object storage, FastAPI, ReportLab/WeasyPrint.

Dependencies. No external API; optional GPS permission.

Prototype recommendation: MUST HAVE.

Feature 2 — Calibrated Font-Height Assistance

Feature & purpose. Estimate height of selected printed characters/numerals only after obtaining a scale reference. Use a printed calibration card/ruler in the same plane as the declaration, perspective correction, and an uncertainty interval.

Coverage. Explicitly addresses readability and font-size requirements.

Differentiation. Most systems will turn pixels into millimetres without scale. We explicitly reject uncalibrated measurement and report: “*Estimated 1.9 mm ± 0.3 mm; physical verification required.*”

Pros / limitations. Directly tackles the hardest requested feature; works well for guided, planar captures. It is unreliable for curved, shiny, crumpled, transparent or severely angled packages.

Risks. Barcode width is not a universal reference because barcode magnification/printing can vary. Use a known-size ArUco/AprilTag reference card or ruler rather than assuming every EAN-13 barcode is 37.29 mm.

Difficulty / feasibility. Medium; feasible with controlled capture.

Legal/regulatory care. Rule 7/schedules and exemptions must be sourced from the latest official consolidated position. Treat estimate as decision-support; authorised physical measurement remains necessary for enforcement action.

Technology. OpenCV (ArUco/AprilTag, homography, quality checks), PaddleOCR, confidence/uncertainty logic.

Dependencies. Printed reference card/ruler; test photos with physical caliper measurements.

Prototype recommendation: MUST HAVE, with controlled scope.

Feature 3 — Batch-Level Compliance History and Change Detection

Feature & purpose. Extract or manually confirm batch/lot code; maintain product-batch history; compare structured declarations and selected visual regions across earlier scans.

Coverage. Product repository, history, retrieval, compliance monitoring and dashboard.

Differentiation. Generic solutions analyse an image in isolation. We show whether label declarations changed from one batch/version to another.

Pros / limitations. Useful for spotting label drift and highly visual in a demo. It does not establish counterfeit status or prove a changed label is unlawful.

Risks. Dot-matrix codes are hard for OCR; promotional or redesign changes can create benign differences. Require manual correction/confirmation of SKU and batch.

Difficulty / feasibility. Medium; highly feasible.

Legal/regulatory care. A change is an inspection lead, not a legal conclusion. Associate each material difference with the specific applicable rule only after officer review.

Technology. PaddleOCR, regex, PostgreSQL/SQLite, normalised declaration records, SSIM/perceptual hash/ORB for limited region comparison.

Dependencies. At least two product/batch records; controlled sample images.

Prototype recommendation: STRONGLY RECOMMENDED.

Feature 4 — Transparent Risk-Based Inspection Prioritisation

Feature & purpose. Create a transparent queue of products/brands/batches that deserve inspection first, using configurable factors such as recent verified potential findings, recurrence, unresolved status and recency.

Coverage. Dashboard, monitoring, repository and inspection history.

Differentiation. Generic dashboards describe the past; this suggests an explainable next action: *"inspect these items first, and here is why."*

Pros / limitations. High operational value and quick to prototype. Production usefulness depends on the quality and volume of authorised historical data.

Risks. Biased/incorrect data can produce unfair targeting. Do not use opaque prediction; show score factors, allow officer override and audit the override.

Difficulty / feasibility. Easy-Medium; feasible.

Legal/regulatory care. Decision-support only, not an enforcement decision. Do not access or scrape official complaint/inspection data without explicit authority. Protect any personal data under DPDP Act, 2023.

Technology. PostgreSQL/SQLite, SQL/Pandas, configurable deterministic score, FastAPI, React/Streamlit.

Dependencies. Historical inspection data; use clearly labelled synthetic records for prototype.

Prototype recommendation: STRONGLY RECOMMENDED.

Baseline Capabilities Required

The four features depend on a small baseline: image upload/camera capture; OCR; extraction of MRP, net quantity, responsible party/address, relevant dates, consumer-care and batch candidates; deterministic, versioned rules; evidence crops; officer confirmation; repository; and PDF/editable export. Use "potential non-compliance" throughout.

Features to Defer or Avoid

Feature	Recommendation	Reason
Live e-commerce scraping/cross-validation	Optional / defer	Good future feature but website terms, dynamic pages, CAPTCHA and anti-bot restrictions can break a live demo. Use authorised/static sample listing data if demonstrated.
Multi-view 3D reconstruction	Avoid now	High risk on reflective/curved packs and likely to reduce prototype reliability.
Counterfeit detection	Avoid now	Needs reliable genuine/counterfeit data and forensic validation; anomaly is not proof.
Blockchain	Avoid	Hashing, audit logs and access controls are simpler and more relevant for prototype.
LLM-driven legal decisions	Avoid	Use deterministic, versioned rules. LLMs can assist search/explanation but must not authoritatively decide compliance.

Prioritised Shortlist

Rank	Feature	Impact, uniqueness and feasibility	Decision
1	Calibrated font-height assistance	Addresses hardest explicit requirement; differentiation is calibration + uncertainty + officer review.	Must Have
2	Evidence-first record	Low effort, high field-workflow credibility and strong reports.	Must Have
3	Batch history/change detection	Longitudinal compliance monitoring is uncommon and demo-friendly.	Strongly Recommended
4	Transparent prioritisation	Turns inspection history into an actionable, explainable queue.	Strongly Recommended
5	E-commerce comparison	Potentially valuable but should only be added after reliable core features and legal/platform review.	Optional

Proposed Positioning Statement

“Rather than another OCR pass/fail tool, our platform helps an officer capture traceable evidence, identify potential declaration and font-size issues using calibrated uncertainty, compare declarations across batches, and focus limited inspections where documented history indicates higher risk.”

Official Resources to Verify Before Coding

- Indian Institute of Legal Metrology, Acts & Rules: <https://iilm.gov.in/more/act-rules>
- Department of Consumer Affairs: <https://consumeraffairs.nic.in/>
- Gazette of India / e-Gazette: <https://egazette.nic.in/>
- MeitY (DPDP Act and privacy resources): <https://www.meity.gov.in/>
- GS1 barcode standards: <https://www.gs1.org/standards/barcodes>
- Official technical/legal verification required: confirm the latest consolidated LMPC Rules, schedules, all amendments, exemptions and commencement dates directly from official notifications before finalising the rule engine.